

HOW TO EXPLAIN THIS PROJECT TO A RECRUITER

Bank Transactions & Account Holders – MySQL Project

Pitch Script + Interview Q&A

This document gives you a ready-to-use spoken script for walking a recruiter or interviewer through this project, followed by answers to the questions you're most likely to be asked about it. Practice the script out loud a few times until it feels natural — don't memorize it word for word, use it as a structure.

1. The 30-Second Version (Elevator Pitch)

Use this when a recruiter asks “tell me about a project on your resume” and you have limited time, such as in a screening call.

“I built an end-to-end SQL analytics project simulating a retail bank's customer and transaction data — about 1,500 accounts and over 12,000 transactions across channels like UPI, NEFT, and ATM. I designed the MySQL schema myself, loaded the data through Workbench, and wrote business queries to answer questions like which branches and channels drive the most transaction value, where customer spending is concentrated, and where operational risks like failed transactions or KYC gaps show up. I documented it the way an analyst actually would — a problem statement first, then the SQL, then an insight report with recommendations.”

2. The Detailed Walkthrough (1.5–2 Minutes)

Use this when the interviewer says “walk me through it” or you're in a technical round with more time. It's structured in five parts — pause naturally between each.

Part 1 – Context

“The idea was to think like a bank's analytics team. They had two raw extracts from the core banking system — an account holders file and a transactions file — and needed someone to turn that into a proper database and a set of business answers, not just raw data.”

Part 2 – Data & Design

“I designed a two-table relational schema: `account_holders` as the customer master, and `bank_transactions` linked to it by a foreign key on `account_id`. I added indexes on the columns I knew I'd filter and join on a lot — `account_id`, `transaction_date`, `transaction_status` — so the queries would run efficiently even with 12,000+ rows. Then I loaded both CSVs into MySQL Workbench using the Table Data Import Wizard, importing accounts first so the foreign key had something to reference.”

Part 3 – Validation Before Analysis

“Before I trusted any insight, I ran exploratory queries — row counts, checking for duplicate primary keys, checking for orphaned transactions with no matching account, and confirming the date range and value ranges made sense. That's a habit I try to carry into every project: don't analyze data you haven't sanity-checked first.”

Part 4 – The Analysis

“Then I wrote 18 business queries grouped into four areas: customer base overview, transaction volume and value, customer and branch-level performance, and operational risk. I kept the SQL intentionally simple — joins, GROUP BY, HAVING, CASE WHEN, a couple of subqueries, and one view — because the goal was clarity and correctness, not showing off syntax.”

Part 5 – What I Found

“A few things stood out. UPI was by far the most-used channel at roughly 30% of all successful transactions, which tracks with how digital payments have shifted in India. About 7% of accounts still had KYC pending, and most of those were still actively transacting — that's a compliance gap I flagged in my report. And Corporate accounts, even though they were under 9% of customers, held disproportionately high balances, which is the kind of thing that would inform where a bank focuses relationship management.”

3. One-Line Answers for Common Follow-Ups

- Tools used: “MySQL Workbench 8.0 for the database, schema design, and all analysis.”
- Data size: “1,500 accounts, just over 12,000 transactions, spanning two years of activity.”
- Deliverables: “A problem statement, four SQL scripts, and a written insight report with recommendations — the same structure I'd use on the job.”
- Time to build: keep this honest and specific to how you actually built it — interviewers respect a clear, factual timeline more than a vague one.

4. Interview Questions & Answers

These are the questions most likely to come up once you mention this project. The answers below are grounded in what you actually built — adapt the wording to sound like you, but keep the substance.

Q1. Why did you choose a banking dataset for this project?

A: Banking data has a good mix of what a real analyst deals with — a customer master table, a high-volume transaction table, categorical fields like channel and status, and natural business questions around revenue, risk, and customer behaviour. It let me practice joins and aggregations on a dataset that actually resembles what I'd see in a corporate analyst role.

Q2. Walk me through your schema design — why two tables and not one?

A: One table would have meant repeating every customer's details — name, city, branch — on every single transaction row, which wastes storage and risks inconsistency if that data ever needs to change. Splitting it into `account_holders` and `bank_transactions`, joined on `account_id`, is basic normalization: each account's details live in exactly one place, and the transaction table stays lean.

Q3. Is this real bank data? Where did it come from?

A: It's a synthetic dataset I generated myself to closely mirror real banking patterns — realistic Indian names, cities, branch structures, transaction channels, and value distributions by customer segment. I'm upfront about that in interviews; using synthetic data for a portfolio project is completely standard, since real customer financial data isn't something you'd ever have legal access to as a training exercise. What matters is that the schema design and SQL are my own work.

Q4. What was the trickiest part of this project?

A: Making the data feel realistic rather than random — for example, making sure account balances actually moved correctly as each transaction happened, and that spend patterns differed sensibly between Retail, Premium, and Corporate customers. On the SQL side, it was deciding what NOT to include — keeping every query minimal and readable instead of stacking unnecessary complexity.

Q5. Why didn't you use window functions or CTEs?

A: I made a deliberate choice to keep this particular project to core SQL — joins, GROUP BY, HAVING, CASE WHEN, and a view — because the audience for it is beginners, and because every one of those 18 queries answers a real business question cleanly without needing more advanced syntax. I do use window functions and CTEs in other projects where the analysis genuinely calls for ranking or running totals.

Q6. How did you make sure the data was clean before analyzing it?

A: I ran a validation pass before touching any business question — checked row counts matched expectations, checked for duplicate primary keys, checked that every transaction's account_id actually existed in the account_holders table using a LEFT JOIN, and confirmed date ranges and value ranges looked sane. That's in my 03_exploratory_data_analysis.sql file.

Q7. What are two or three key insights from your analysis?

A: UPI drives about 30% of all transaction volume by count, which shows how dominant digital payments have become. Around 7% of accounts have KYC still pending, and most of those are actively transacting, which is a compliance gap worth flagging. And a small group of Corporate customers holds a disproportionately large share of account value, which would justify dedicated relationship management.

Q8. How would this project change if the data was 10x or 100x larger?

A: The schema wouldn't need to change, but I'd lean more on the indexes I already added, consider partitioning the transactions table by date, and move from ad-hoc queries to the view I built — or a small set of summary/aggregate tables — so dashboards querying this data don't re-scan millions of rows every time.

Q9. What would you add if you extended this project further?

A: I'd add a simple dashboard on top of the view I created — in Power BI or Tableau — so the insights update automatically as new data comes in, and I'd add a few window-function queries like month-over-month growth or a running balance check, since those are common asks in real analyst roles.

Q10. How does this project relate to the role you're applying for?

A: It demonstrates the full analyst workflow end to end — understanding a business problem, designing a schema, loading and validating data, writing SQL that answers specific business questions, and communicating findings in a report a non-technical stakeholder could act on. That workflow is the same regardless of domain, which is why I built it to look like a real deliverable rather than just a set of queries.

5. A Quick Reminder Before You Walk In

Say the numbers with confidence — you generated this data and wrote every query, so you know it better than anyone in the room. If you're asked something you genuinely don't know, it's fine to say “I'd need to check that” rather than guessing; interviewers notice honesty more than they notice a perfect answer.

Prepared by: Aishwarya Mate